

PERSONAL
INFORMATION**Jinho Chang**

Seoul, South Korea

Personal Webpage <https://github.com/jinhojsk515>**in** www.linkedin.com/in/jinho-chang-09a2a1284**Google Scholar** <https://scholar.google.com/citations?user=GrHZicUAAAAJ>

Ph.D. student at KAIST, currently a member of the BioImaging-Signal Processing & Learning Lab(BISPL). I am working on diffusion and flow-matching models for their variational applications and inference-time techniques, and multi-modal models for informative representations and cross-modal generation. Also, leveraging my chemistry background, I aim to apply state-of-the-art AI techniques to various important chemical challenges with minimal adaptation costs.

RESEARCH
INTERESTS

- Inference-Time Scaling** : Exploiting analytic and variational analyses on diffusion and flow-matching models towards training-free control of the sampling process, including reward-guided editing, negative guidance, score distillation, molecule optimization, etc.
- Alignment of Generative Models** : Since training generative models for specific desirable features from scratch is costly and often limited, we establish a set of reward-alignment frameworks that exploit various aspects(e.g., initial noise, intermediate guidance, stochasticity, etc.) of the base models towards task-specific external objectives while preserving prior knowledge.
- AI for Chemistry** : Assessing molecular tasks that are hard to solve in principle, including conditional molecule design and editing, synthesizability and retro-synthesis prediction, intrinsic property prediction, etc.
- Multimodal Learning** : Leveraging various multimodal learning of a shared data space or representation alignment, for better cross-modal understanding and generation on medical imaging and molecular structure and its properties.

SELECTIVE
PUBLICATIONS

The full list with more details can be found in [Google Scholar Page](#).

- **Reward Score Matching : Unifying Reward-based Fine-tuning for Flow and Diffusion Models** [[link](#)] [[Alignment](#)]
Jeongjae Lee, Jinho Chang*, Jeongsol Kim, and Jong Chul Ye* *ICML SPIGM (Best Paper), 2026*
We unify reward-based fine-tuning methods for diffusion and flow models under reward score matching (RSM), which casts alignment as value-guided score matching. This view isolates key design choices - value guidance estimation and timestep-wise optimization strength - clarifying tradeoffs and enabling simpler, more efficient variants across differentiable and black-box tasks.
- **Contrastive CFG : Improving CFG in Diffusion Models by Contrasting Positive and Negative Concepts** [[link](#)] [[Alignment](#)] [[Inference-Time Scaling](#)]
Jinho Chang, Changsun Lee, Hyungjin Chung, and Jong Chul Ye *ICML, 2026*
We point out the drawback of naively negating the Classifier-Free Guidance (CFG) term, and propose a novel approach to enhance negative CFG in conditional diffusion models using contrastive loss. Our method overcomes the distortions caused by traditional negative CFG, achieving superior sample quality and effective removal of unwanted features across various scenarios.
- **Training-Free Reward-Guided Image Editing via Trajectory Optimal Control** [[link](#)] [[Alignment](#)] [[Inference-Time Scaling](#)]
Jinho Chang, Jaemin Kim*, and Jong Chul Ye* *ICLR, 2026*
Prior reward-guided sampling in diffusion and flow-matching has been studied primarily for generation and remains less explored for image editing. We introduce a training-free editing framework that formulates reward-guided editing as a trajectory optimal control problem to drive the endpoint toward higher reward. Our method outperforms inversion-based training-free baselines, achieving higher rewards while preserving source fidelity without reward hacking.

- **LDMol : Text-Conditioned Molecule Diffusion Model Leveraging Chemically Informative Latent Space** [link] **AI4Chem** **Multimodal**
Jinho Chang and Jong Chul Ye *ICML, 2025*
 We present a new latent diffusion model LDMol for molecule generation with complex conditions like natural texts. Leveraging a chemically informative latent space obtained with contrastive learning, LDMol not only outperforms previous text-to-molecule generative models, but can also be applied to downstream tasks like retrieval and molecule editing.
- **DreamMotion : Space-Time Self-Similarity Score Distillation for Zero-Shot Video Editing** [link] **Inference-Time Scaling**
Hyeonho Jeong, Jinho Chang, Geon Yeong Park, and Jong Chul Ye *ECCV, 2024*
 We present a score distillation sampling for video editing to circumvent the standard reverse diffusion process. We propose to match the space-time self-similarities during the score distillation, demonstrating its superiority in altering appearances while accurately preserving the original structure and motion.
- **LLM-CXR : Instruction-Finetuned LLM for CXR Image Understanding and Generation** [link] **Multimodal**
Suhyeon Lee, Wonjun Kim*, Jinho Chang, and Jong Chul Ye* *ICLR, 2024*
 Taking inspiration from previous work on the transformer and VQ-GAN combination for bidirectional image and text generation, we developed a method for instruction-tuning a text-only LLM to gain vision-language capabilities for medical images. We let LLM understand tokenized visual inputs by instructing it to answer questions about image inputs and generate image outputs.
- **Bidirectional Generation of Structure and Properties Through a Single Molecular Foundation Model** [link] **AI4Chem** **Multimodal**
Jinho Chang and Jong Chul Ye *Nature Communications, 2024*
 We introduce a multimodal molecular pre-trained model that integrates molecular structures and their biochemical properties, which enables various multimodal and unimodal tasks like conditional molecule generation, property prediction, molecule classification, and reaction prediction.
- **Drug-likeness Scoring Based on Unsupervised Learning** [link] **AI4Chem**
Jinho Chang, Kyunghoon Lee, Seonghwan Seo, and Woo Youn Kim *Chemical Science, 2022*
 We proposed a novel unsupervised learning model that could quantify the drug-likeness of a given molecule, by adopting a language model trained via unsupervised learning. Our model showed more consistent performance across different datasets and gave more reasonable scores.

REVIEWER PARTICIPATION

<i>Nature Communications</i>	2024 -
<i>Communications Chemistry</i>	2024 -
<i>International Conference on Learning Representations</i>	2025, 2026
<i>Conference on Computer Vision and Pattern Recognition</i>	2025
<i>International Conference on Computer Vision</i>	2025
<i>International Conference on Machine Learning</i>	2026
<i>Neural Information Processing Systems</i>	2026

EXPERIENCES

Internship as AI developer @ HITS (https://hits.ai/)	January 2021 - June 2021
I participated in the development of a scaffold-based molecular generative model via reinforcement learning for compounds that are both synthesizable and possess specific pharmaceutical properties.	
Lab Manager @ BISPL (https://bispl-website.github.io/)	2026
I served as the laboratory manager, overseeing overall laboratory resources and responsible research proposals, coordinating laboratory members, etc.	

REWARDS & HONORS

National Science & Technology Scholarship	March 2019 - August 2022
KAIST Presidential Fellowship (KPF)	March 2018 - August 2022

EDUCATION

Ph.D. student in Artificial Intelligence Korea Advanced Institute of Science and Technology (KAIST)	September 2022 - August 2028
B.S. in Computer Science and Chemistry (double major) Korea Advanced Institute of Science and Technology (KAIST)	March 2018 - August 2022